COURSE STRUCTURE FOR B.TECH. Mechanical Engineering SECOND YEAR SEM – IV (w. e. f. 2024-25)

SEMESTER-IV (Subjects)				B.TECH. MECHANICAL ENGINEERING Sem-IV											
Sr. No	Category Code	Course Code	Course Name	Teaching Scheme					Exam Scheme						
									Theory			Practical		Total	
				L	T	P	C	Hrs/wk	MS	ES	CE	LE	LE/Viva	Marks	
1	OE	24ME221T	Renewable Energy	3	0	0	3	3	25	25	50	-	-	100	
2	IND	24ME205T	Industry 4.0	2	0	0	2	2	25	25	50	-	-	100	
3	IND	24ME205P	Industry 4.0 Laboratory	0	0	2	1	2	-	-	-	25	25	50	
4	PCC	24ME206T	Fluid Mechanics	3	0	0	3	3	25	25	50	-	-	100	
5	PCC	24ME206P	Fluid Mechanics Laboratory	0	0	2	1	2	-	-	-	25	25	50	
6	PCC	24ME207T	Design and Kinematics of Machines	3	0	0	3	3	25	25	50	-	-	100	
7	PCC	24ME207P	Design and Kinematics of Machines Laboratory	0	0	2	1	2	-	-	-	25	25	50	
8	PCC	24ME208T	Engineering Metallurgy	3	0	0	3	3	25	25	50	-	-	100	
9	PCC	24ME208P	Engineering Metallurgy Laboratory	0	0	2	1	2	-	-	-	25	25	50	
10	PCC	24ME209T	Strength of Material	3	0	0	3	3	25	25	50	-	-	100	
11	PCC	24ME209P	Strength of Material Laboratory	0	0	2	1	2	-	-	-	25	25	50	
12	NCC	24INT251	Industrial Orientation	0	0	0	0	0	-	-	-	-	-	-	
			Total	**17**	**0**	**10**	**22**	**27**							

MS = Mid Semester ES = End Semester CE = Continuous Evaluation LW = Laboratory work LE = Laboratory Exam

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME205T					Industry 4.0 for Mechanical Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	0	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. To Understand the Foundations and Architectural Principles of Industry 4.0.
2. To Explore the Internet of Things (IoT) and Data Analytics in Industrial Applications.
3. To Gain Insight for Digital Twin and Virtual Prototyping Technologies.
4. To Apply Advanced Automation and Robotics within Industry 4.0 Frameworks

UNIT I: INTRODUCTION TO INDUSTRY 4.0 **10Hrs.**

Introduction to Industry 4.0, Definition, General framework, Application areas, Introduction to the Evolution of Industrial revolutions, Key features, Need and benefits, Introduction to Industry 4.0 core technologies: Big data, Advanced Robotics, Simulation, Integration, Internet of Things (IoT), Artificial Intelligence (AI), Cybersecurity, Cloud computing, Additive manufacturing and Augmented Reality. Overview of machine learning for Industry 4.0. Transformation of industrial processes through the integration of modern technologies such as sensing and actuation, communication, and computational processing.

UNIT II: THE INTERNET OF THINGS AND DATA ANALYTICS **10Hrs.**

Fundamentals of IoT and its components, Sensor Technology and Data Acquisition in IoT-enabled Systems, Types and Functions of Sensors: temperature, pressure, vibration, gas and flow sensors. Connectivity solutions for industrial applications. Techniques for complex data analysis in mechanical engineering. Fundamentals of Image Processing and Object detections.

UNIT III: DIGITAL TWINS AND VIRTUAL PROTOTYPING **10Hrs.**

Basics of Digital Twins, Building Digital Twins, Integration with IoT, digital twins for predictive maintenance and operational efficiency, Fundamentals of Augmented and Virtual Reality, Augmented and Virtual Reality in design, assembly, and maintenance. Developing AR/VR Applications, Integration with real time applications, Future Trends.

UNIT IV: ADVANCED AUTOMATION, ROBOTICS, AND FUTURE TECHNOLOGIES **10Hrs.**

Automation in Supply Chain and Logistics, Automated warehousing and inventory management, Human-Robot Collaboration, Edge computing vs. cloud computing in industrial applications, Integrating IoT devices with digital twins through cloud services. Real-world applications focusing on automation and smart manufacturing.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Remember and Understand Industry 4.0 Principles
CO2 : Apply IoT and Data Analytics in Industrial Systems.
CO3 : Analyze Sensor Data for Mechanical Engineering Applications.
CO4 : Design Digital Twins and Virtual Prototypes.
CO5 : Evaluate Advanced Automation and Robotics Solutions.
CO6 : Synthesize Knowledge to Innovate in Smart Manufacturing.

TEXT/REFERENCE BOOKS

1. K. Palanikumar et al.,"**Machine Intelligence in Mechanical Engineering** ", Elsevier.
2. A. Gilchrist, "**Industry 4.0: The Industrial Internet of Things**", APress.

S. Mishra et al., "**Introduction to Industrial Internet of Things and Industry 4.0**", Routledge

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME205P					Industry 4.0 Laboratory for Mechanical Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. Equip students with foundational programming skills in Python and Open CV.
2. To learn the architectures of Arduino and Raspberry Pi platforms.
3. Provide hands-on experience with Arduino and Raspberry Pi platforms.
4. To apply Industry 4.0 architectures in real industrial applications.

LIST OF EXPERIMENTS

1. Introduction to Python and Machine Learning Libraries.
2. To understand the principles of linear regression and its application in predictive modelling.
3. Introduction to Computer Vision with OpenCV.
4. To utilize Pre-trained Models for object detection.
5. Introduction to Arduino and its architectures.
6. To design and develop a line follower autonomous vehicle model.
7. To monitor and display pressure values in a pneumatic system.
8. Introduction to Raspberry Pi and its architectures.
9. To develop an environmental monitoring system in manufacturing industry.
10. Project 1 submission using Open CV.
11. Project 2 submission using Python.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the principles and applications of Python programming and Open CV.
CO2 : Apply machine learning models for industrial applications.
CO3 : To understand and utilize Arduino and Raspberry Pi for various applications.
CO4 : Demonstrate the ability to integrate sensors with Arduino and Raspberry Pi.
CO5 : To identify and apply sensors in various industrial settings.
CO6 : Evaluate the performance and reliability of sensor-integrated systems.

TEXT/REFERENCE BOOKS

1. H Jake VanderPlas, "**Python Data Science Handbook: Essential Tools for Working with Data**", O'Reilly Media.
2. Adrian Kaehler and Gary Bradski, "**Learning OpenCV 3: Computer Vision in C++ with the OpenCV Library**", O'Reilly Media.
3. Simon Monk, "**Programming the Raspberry Pi: Getting Started with Python**", McGraw-Hill Education.
4. Massimo Banzi and Michael Shiloh, "**Getting Started with Arduino: The Open-Source Electronics Prototyping Platform**", Maker Media, Inc.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME206T					Fluid Mechanics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To develop the basic understanding of fluid static, kinematics and dynamics.
2. To develop understanding to apply continuity, energy and momentum equations to engineering problems involving flow measurement and energy losses.
3. To develop analytical skills to deal with various types of fluid flow problems.

UNIT 1 FLUID STATIC AND KINEMATICS 10Hrs.

Fluid Statics: Properties of fluids, Archimedes Principle, Buoyancy and stability of floating and submerged bodies, Meta-centric height. Manometers. Kinematics: Types of flow, Concept of continuum, Pressure and stress tensor. **Fluid Kinematics:** Lagrangian and Eulerian description, Streamline, Streakline and Pathline, Acceleration of a fluid element, Continuity equation, Stream function, Rotation and angular deformation, Irrotational flow, Velocity potential.

UNIT 2 FLUID DYNAMICS 11Hrs.

Integral Flow Analysis: Reynolds transport theorem, Conservation of mass/continuity equation, Conservation of linear and angular momentum for a control volume in inertial and accelerating reference frames, Energy equation, Bernoulli's equation and engineering applications. **Differential Analysis of Flow:** Stokes law of viscosity and Navier-Stokes equations.

UNIT 3 VISCOUS FLOW 11Hrs.

Reynolds experiment: laminar and turbulent flow. Plane Poiseuille flow, Couette flow, Hagen-Poiseuille flow; Friction factor and Moody's diagram, Losses in pipes, Pipe fittings and losses.

UNIT 4 DIMENSIONAL ANALYSIS AND BOUNDARY-LAYER THEORY 10Hrs.

Dimensional Analysis and Similitude: Buckingham-PI theorem, Non-dimensional parameters, Problem-solving using non-dimensionalization; Similitude and Non-dimensional numbers.

Boundary-Layer Theory: Boundary layer development and Boundary layer thicknesses, Characteristics of Boundary Layer, Boundary Layer Equations. Flow around bluff bodies, development of drag and lift on aerofoil.

Max. 42 Hrs.

COURSE OUTCOMES:

On completion of the course, student will be able to:

CO1: Understand the properties of fluid and apply the laws of fluid statics to determine buoyancy and stability of floating and submerged bodies.
CO2: Evaluate transport properties of fluid for internal and external flows.
CO3: Apply conservation principles of mass, linear momentum, and energy to fluid flow systems to determine flow quantities and energy losses in fluid flow systems.
CO4: Evaluate and arrive at reasonable approximations for a fluid flow problem where flow is governed by the continuity equation and Navier-Stokes equation.
CO5: Design simple pipe systems to deliver fluids under specified conditions.
CO6: Apply dimensional analysis to predict physical parameters that influence the flow in fluid mechanics.

TEXT/REFERENCE BOOKS

1. F. White, **"Fluid Mechanics",** Tata-McGraw Hill publishers, 2011.
2. Y. A. Cengel and J. M. Cimbala, **"Fluid Mechanics Fundamentals and Applications",** Tata-McGraw Hill Publishers, 2015.
3. R. Fox and A. McDonald, **"Introduction to Fluid Mechanics",** John Wiley Publishers, 2011.
4. V. Streeter and Benjamin, **"Fluid Mechanics: First SI-Metric",** McGraw-Hill, Auckland 2001.
5. Currie, I.G., **"Fundamental Mechanics of Fluids",** 4th Edition, CRC Press, 2012.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME206P					Fluid Mechanics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart knowledge on construction and working principle of various flow measuring devices.
2. To impart practical exposure on the performance evaluation methods of various flow measuring equipment and energy losses fluid flow through pipe.
3. To develop understanding on experimental data analysis, technical report writing and work in teams.

LIST OF EXPERIMENTS

Experiment 1: Measurement of viscosity

Experiment 2: Verification of Bernoulli's principle

Experiment 3: Discharge coefficient of Venturimeter

Experiment 4: Discharge coefficient of Orificemeter

Experiment 5: Calibration of Rotameter

Experiment 6: Reynolds apparatus

Experiment 7: Flow measurement with Pitot tube

Experiment 8: Friction losses in circular pipe

Experiment 9: Effect of pipe diameter and length on friction losses in pipe

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Apply mass and energy conservation principles to fluid flow systems to estimate energy losses.
CO2: Evaluate the coefficient of discharge and calibration of flow measuring devices.
CO3: Identify flow patterns and flow types for internal flow systems.
CO4: Evaluate, compare and contrast experimental results of major and minor losses for internal flow systems with theoretical trends, and to attribute observed discrepancies to various errors.
CO5: Design and conduct experiments, analyze and interpret data for fluid flow systems.
CO6: Compile and interpret the experimental data under various operation conditions.

TEXT/REFERENCE BOOKS

1. F. White, **"Fluid Mechanics",** Tata-McGraw Hill publishers, 2011.
2. Y. A. Cengel and J. M. Cimbala, **"Fluid Mechanics Fundamentals and Applications",** Tata-McGraw Hill Publishers, 2015.
3. C.S.P. Ojha, R.Berndtsson, P.N. Chandramouli, **"Fluid Mechanics and Machinery",** Oxford University Press, 2012

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME207T					Design and Kinematics of Machines					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn the fundamentals of design and kinematics.
2. To comprehend joint and spring design principles for different mechanical applications.
3. To acquire proficiency in designing shafts, keys, and couplings to ensure strength and rigidity.
4. To apply kinematic synthesis techniques for creating linkages and achieving precise motion control in mechanisms

UNIT I: INTRODUCTION TO DESIGN, KINETICS AND KINEMATICS 10Hrs.

Introduction: Design procedure, coding of materials, factors of safety, standards, aesthetics, ergonomics, design aspects in selection of materials and in manufacturing, fundamentals of kinematics and mechanisms, kinematic link, types, kinematic pair, types of motions, kinematic pairs and chain, types of joints, mechanism, machine, degree of freedom (mobility), inversions of simple mechanisms, Grashoff's law, role of design in industry 4.0, concurrent engineering, CAE, CAD, simulation, various computation tools used to design, modelling and analysis

UNIT II: DESIGN OF JOINTS AND SPRING 10Hrs.

Design of cotter joint and knuckle joint, Design riveted and welded joints: Terminology of a riveted joint, types of rivets and riveted joints, design of a riveted joint, joint efficiency, eccentrically loaded riveted joint, types of welded joints, stresses in welded joints and design of it. Design of mechanical springs: Types of springs, materials for springs, stresses, deflection and buckling of helical springs, helical springs of non-circular cross-sections, helical tension spring, energy storage capacity, helical torsion springs, co-axial springs, leaf springs

UNIT III: DESIGN OF SHAFTS, KEYS AND COUPLINGS 10Hrs.

Design of Shafts, Keys and Couplings: Shaft Material, Design of solid and hollow shafts for strength and rigidity, Design of shafts for combined bending and axial loads, Types of couplings, Design of flange coupling, Design of keys

UNIT IV: KINEMATIC SYNTHESIS 10Hrs.

Synthesis of Linkages: Type, number and dimensional synthesis. Path, function and motion generation (Body guidance). Precision Positions, Chebyshev spacing, Crank Rocker mechanisms, Analytical synthesis: Derivation of Freudenstein's equation, Three-position function generation using Freudenstein's equation.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : **Recall** the fundamentals of design and the theory of kinematics and mechanisms.
CO2 : **Explain** the various mechanical components such as rivets, welded joints, shafts, couplings, springs and synthesis of mechanisms.
CO3 : **Apply** the various mechanical design and kinematics principles to design various mechanical components.
CO4 : **Analyse** the position of machines and their components.
CO5 : **Estimate** the dimensions and position of various mechanical components.
CO6 : **Formulate and design** rivets, welded joints, shafts, couplings, springs and various mechanisms.

TEXT/REFERENCE BOOKS

1. V. Bhandari, **"Machine Design"**, Tata-McGraw Hill Publishers.
2. J. Shigley, C. Mischke, R. Budynas, **"Mechanical Engineering Design"**, Tata-McGraw Hill Publishers.
3. S. S. Ratan, **"Theory of Machines"**, Tata Mc Graw-Hill publications, New Delhi
4. R. Norton, **"Machine Design: An Integrated Approach"**, Pearson Education Publishers.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME207P					Design and Kinematics of Machines Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. Learn to design power screws and levers effectively.
2. Analyze and design cam profiles for desired motion control
3. Develop proficiency in graphical analysis techniques for simple mechanisms.
4. Apply synthesis methods to achieve specific motion requirements in mechanisms.

LIST OF EXPERIMENTS

1. Design and drawing of power screw
2. Design and drawing of lever
3. Design and analysis of cam profile through follower displacement diagrams
4. Analysis and drawing of simple mechanisms by relative velocity method
5. Analysis and drawing of simple mechanisms by instantaneous center method
6. Analysis and drawing simple mechanisms by acceleration diagram and coriolis component of acceleration,
7. Graphical synthesis: two and three position synthesis using relative pole method
8. Graphical synthesis: inversion method for single slider crank and four bar mechanism,
9. Three position motion synthesis of four bar mechanism.
10. Development of prototype of simple mechanism

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : **Recall** the fundamentals of design and the theory of kinematics and mechanisms.
CO2 : **Explain** the various mechanical components such as lever, power screws, cams and velocity-accelerations of mechanisms and synthesis of mechanisms.
CO3 : **Apply** the various mechanical design and kinematics principles to design various mechanical components.
CO4 : **Analyse** the design of motion of machines and their components.
CO5 : **Estimate** the dimensions, velocity and acceleration of various mechanical components.
CO6 : **Formulate and design** cams, power screws and various mechanisms.

TEXT/REFERENCE BOOKS

1. V. Bhandari, **"Machine Design"**, Tata-McGraw Hill Publishers.
2. J. Shigley, C. Mischke, R. Budynas, **"Mechanical Engineering Design"**, Tata-McGraw Hill Publishers.
3. S. S. Ratan, **"Theory of Machines"**, Tata Mc Graw-Hill publications, New Delhi
4. R. Norton, **"Machine Design: An Integrated Approach"**, Pearson Education Publishers.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME208T					Engineering Metallurgy					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	-	-	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To know the importance of metallurgy in manufacturing.
2. To learn different types of mechanical properties and their evaluations.
3. To study microstructural features in metallic materials.
4. To acquire knowledge of different types of metallic materials.

UNIT 1 Testing of metals **9Hrs.**

Concept of stress and strain, strength, elasticity, plasticity, stiffness, resilience, toughness, malleability, ductility, and brittleness. Mechanical testing: tensile, compression, hardness, fatigue, wear, creep, impact. NDT: visual inspection, dye penetration test, radiography, edy current, ultrasonic. DBT curve. ASTM standards for mechanical testing.

UNIT 2 Structure Property Relation **10 Hrs**

Engineering metals and alloys, Elastic and plastic deformation, deformation in a single crystal and polycrystalline metal, Critical resolved shear stress, plastic deformation mechanisms-slip and twin, effect of defects on deformation mechanism, work hardening, fracture in metals, changes in properties due to deformation, re-crystallization, cold working and hard working. Metallography: Optical and electron microscopy (OM, SEM, EBSD, TEM, XRD).

UNIT 3 Ferrous and Non-Ferrous materials **10 Hrs.**

Ferrous metals and Designation: Wrought and cast components, allotropy of iron, iron-carbon diagram, plain carbon steels, limitations of plain carbon steel, and advantages of alloy steels. Effect of alloying elements on mechanical properties of steel, alloy steels, tool steels, stainless steels, cast irons. Designation of steels and cast iron. Corrosion in metallic materials. Non- ferrous metals and alloys: Al, Cu, Mg, Ti, and Ni.

UNIT 4 Heat treatment and advanced materials **10 Hrs.**

Effect of non- equilibrium cooling on microstructure and properties of steel, TTT diagram for 0.8% carbon steel, Isothermal treatments, continuous cooling transformation curves, critical cooling rate & heat treatments like annealing, normalizing, hardening and tempering. Hardenability of steels, jominey end quench test, surface hardening treatment scarburizing. Nitriding, carbonitriding, tufftride, sursulf, induction hardening and flame hardening. Advanced materials: types and properties of composite materials, high temperature materials, cryogenic materials, shape memory alloy.

Total 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 Define mechanical properties of materials.
CO2: Understand fundamental understanding of Mechanical, Metallurgical and NDT testing.
CO3: Distinguish Ferrous and non-ferrous materials.
CO4: Evaluate relation between structure and properties.
CO5: Analyse importance of the heat treatment processes.
CO6: Remember ASTM Standards and its relevance for mechanical testing

TEXT/REFERENCE BOOKS

1. William D Callister, Jr., Materials Science and Engineering, Wiley India (P) Ltd.
2. F.C. Compbell, Elements of Metallurgy and Engineering Alloys, ASM International, Ohio.
3. George Dieter, Mechanical metallurgy, McGraw-Hill.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME208P					Engineering Metallurgy Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
-	-	2	1	2	-	-	-	25	25	50

COURSE OBJECTIVES

1. To learn the practical significance of different types of mechanical testing of materials and their interpretation
2. To develop an understanding on the methods followed for visualizing the microstructures and constituent phases of the materials, their analysis and their impact on the observed properties
3. To study the materials and their properties while correlating their microstructures and defects with different types of manufacturing processes and their parameters

UNIT 1 Mechanical Testing of Materials

Familiarize with ASTM Standards - E6, E8, E21, E23, E92, E190, E290, and G65

1. Generate both Engineering and True Stress Vs Strain curves and measure the tensile properties of the materials including modulus of resilience, and interpret the importance of each property
2. Correlate the tensile strength with hardness data and understand the variability of material properties.
3. Hardness measurements - Micro and Macro, including Knoop hardness test, and study the variations in micro and macro hardness of the welded steel samples using Vickers hardness test.
4. Correlate the Impact energy to the temperate and understand the importance of Ductile-to-Brittle Transition behaviour of the materials
5. Develop S-N curves for steels and determine their fatigue life & fatigue strength

UNIT 2 Metallography and Property Correlation

Familiarize with ASTM Standards – E340 and E527

1. Over view of Metallography procedures, and Metallurgical microscopes their construction, applications and limitations
2. Overview of Scanning Electron Microscopy and its advantages over optical microscopy
3. Metallographic examination of steels and alloys, their phase analysis, application of lever rule in phase analysis and correlating to the phase diagrams
4. Number Familiarize with ASTM Grain size chart and determine the ASTM Grain Size
5. Metallography of weldments - Macro and Microstructures
6. Metallographic examination of Ferrous Metals including Stainless Steels
7. Metallographic examination of Non Ferrous Metals - Aluminium and Copper
8. Fractrographay of different materials (Tensile, Brittle, Fatigue and Creep)
9. Wear Testing of Metallic Samples (ASTM G65)
10. Super plasticity of metallic materials and elevated temperature tensile tests of metallic materials (ASTM E 21)
11. In-situ NDT tool for health assessment, and practical demonstration of various NDT methods
12. Effect of heat treatments on microstructures

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the working principles and operations of various instruments.
CO2 - Analyse the effect of different parameters on the measurements and methodologies.
CO3 - Demonstrate the significance of different instruments for different measurements.
CO4 - Judge the uses of different techniques for different applications in practical life.
CO5 - Appraise the limitations of each technique and methodologies followed.
CO6 - Case study the materials and correlate to their properties

TEXT/REFERENCE BOOKS

1. Donald C. Zipperian, METALLOGRAPHIC HANDBOOK, PACE Technologies, USA
2. William D. Callister, Jr. Materials Science and Engineering an Introduction
3. Various ASTM standards and Lab Manual - PDPU

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME209T					Strength of Material					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	-	-	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart the knowledge of fundamental concepts of stresses and strains.
2. To obtain the analytical and graphical solutions of principle stress and strain.
3. To get acquainted with the theories on flexural stresses and bean deflections.
4. To get accustomed to the torsional forces in solid and hollow shafts.

UNIT I: STRESS, STRAIN AND STRESS TRANSFORMATION 9 Hrs.

Stress, strain, types, stress-strain diagram, practical example of failures associated with stress/strain, role of stress/strain/failures in mechanical design, factor of safety, statically indeterminate bars. Principal stresses and strains: stresses on an inclined section of a bar under axial loading, compound stresses: normal and tangential stresses on an inclined plane for biaxial stresses. Two perpendicular normal stresses accompanied by a state of simple shear, Stress transformation, Mohr's circle stresses- Principle stresses, maximum shear stresses and strains- analytical and graphical solutions.

UNIT II: FLEXURAL STRESSES IN BEAMS 10 Hrs.

Shear force and bending moment by taking examples of S.F and B.M diagrams for beams subjected to concentrated loads, uniformly distributed loads (U.D.L) and uniformly varying loads (U.V.L.) and their combinations, Point of contra flexure.Flexural stresses: Theory of simple bending, Assumptions, Derivation of bending equation (moment-curvature equation), Neutral axis, Determination bending stresses (flexural formulae), section modulus of rectangular and circular sections (Solid and Hollow), T, I, Angle, Channel sections, Design of simple beam sections.

UNIT III: SHEAR STRESS IN BEAMS AND TORSION OF SHAFT 10 Hrs.

Shear Stress: Shear stress variation, applicability to beam, shear stress distribution for various sections such as rectangle, solid circle, hollow circle, T section, I section. Torsion: Introduction torsion, comparison between solid and hollow shaft with regard to their strength and weight, Shaft in parallel and series.

UNIT IV: DEFLECTION OF BEAMS AND ENERGY METHODS 11 Hrs.

Bending into a circular arc, slope, deflection and radius of curvature, Differential equation for the elastic line of a beam, Determination of slope and deflection using Macaulay's methods. Energy Methods: Strain energy, Strain energy for Normal, shear and bending stress, Castigliano's theorm.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Define stresses, strains and stress transformation.
CO2 : Explain slope and deflection of beam.
CO3 : Apply the pure bending and pure twisting theories to beam and shaft respectively.
CO4 : Analyse the shear force and stress, and bending moment and stress diagrams of beams.
CO5 : Estimate the bending and shear stresses of beams.
CO6 : Evaluate deflection in beams and torsional stresses in shaft.

TEXT/REFERENCE BOOKS

1. James M Gere, **"Mechanics of Materials"**, Cengage publication.
2. Beer and Johnston, **"Mechanics of Materials"**, Tata Mc Graw hill..
3. R. C. Hibbeler, **"Mechanics of Materials"**, Prentice Hall, Pearson, India.
4. S. S. Ratan, **"Strength of Materials"**, Tata Mc Graw hill.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – IV

24ME209P					Strength of Materials Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To gain the fundamental knowledge of engineering in finding the strength of mechanical element subjected to various type of loadings.
2. To identify, formulate and solve the engineering problems exist in mechanical element under damage.
3. To understand the failure behavior of structural element and impact of usage of material in various application.
4. To learn about the physical aspect related to hardness, toughness, stress, impact, fatigue, tension, compression, torsion and fatigue.

LIST OF EXPERIMENTS

1. To study the stress-strain characteristics of mild steel, aluminum by conducting tensile test on UTM.
2. To determine the Young's modulus of the material by conducting deflection test on a simply supported beam.
3. To determine the Modulus of rigidity by conducting Torsion test on Solid shaft.
4. To find the Brinnell's hardness numbers of (a) Steel (b) Brass (c) Aluminum (d) Copper by conducting hardness test.
5. To find the Rockwell hardness numbers of (a) Steel (b) Brass (c) Aluminum (d) Copper by conducting hardness test.
6. To find compressive strength of wood and concrete by conducting compression test.
7. To find impact strength of (a) steel (b) aluminum by conducting Izod impact test.
8. To find impact strength of (a) steel (b) aluminum by conducting Charpy impact test.
9. To study and demonstrate the stress field using digital image correlation system.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Illustrate the behaviour of material under impact condition.
CO2 : Understand the deflection of different sections at different loading conditions.
CO3 : Sketch stress- strain curve of ductile material under tensile loading.
CO4 : Compare compression strength between wood and concrete.
CO5 : Estimate the hardness of metals.
CO6 : Evaluate elastic constants using flexural and torsion test.

TEXT/REFERENCE BOOKS

1. James M Gere, **"Mechanics of Materials"**, Cengage publication.
2. Beer and Johnston, **"Mechanics of Materials"**, Tata Mc Graw hill..
3. R. C. Hibbeler, **"Mechanics of Materials"**, Prentice Hall, Pearson, India.
4. S. S. Ratan, **"Strength of Materials"**, Tata Mc Graw hill.
5. R.K.Rajput, **"Strength of Materials"**, S.Chand & Co, New Delhi.